Project Initialization and Planning Phase

Date	04 July 2026 – 05 July 2026
Team ID	RW-2026-01
Project Title	Rising Waters – AI-Based Flood Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The **Rising Waters** project aims to enhance flood prediction and disaster preparedness using Artificial Intelligence and Machine Learning. The system analyses historical flood records, rainfall patterns, weather conditions, and river water levels to accurately predict flood risks and provide early warnings. By delivering timely alerts to citizens and disaster management authorities, the solution helps minimize loss of life, property damage, and environmental impact while supporting informed decision-making during flood emergencies.

Project Overview:

Objective	The primary objective of Rising Waters is to develop an AI-powered flood prediction system that accurately forecasts flood risks using machine learning and real-time environmental data, enabling early warning and effective disaster management.
Scope	The project includes collecting and preprocessing weather and historical flood data, training an XGBoost machine learning model, predicting flood risks, generating real-time alerts, and providing an interactive dashboard for users and disaster management authorities.

Problem Statement:

Description	Floods have become increasingly frequent due to climate change, heavy rainfall, poor drainage systems, and rapid urbanization. Existing warning systems often provide delayed or inaccurate predictions, leading to loss of life, property damage, and economic losses.

Impact	Implementing an AI-based flood prediction system will improve prediction accuracy, provide early flood warnings, support emergency response planning, reduce disaster-related losses, and enhance public safety.

Proposed Solution:

Approach	Develop a machine learning-based flood prediction system using the XGBoost algorithm. The system processes historical flood data and real-time weather information to predict flood risks and automatically generate alerts.
Key Features	• AI-powered flood prediction using XGBoost. • Real-time weather and river water level monitoring. • Automated flood risk assessment. • Early warning and alert notification system. • Interactive dashboard for prediction results and reports. • Historical flood data analysis and visualization. • Scalable and user-friendly web application.

Resource Requirements:

Resource Type	Description	Specification/Allocation
Hardware	CPU/GPU specifications, number of cores	Intel Core i5/Ryzen 5 Processor or above
	RAM specifications	Minimum 8 GB RAM
	Disk space for data, models, and logs	Minimum 256 GB SSD (Recommended 512 GB SSD)

Software	Backend Framework	Python, Flask
	Machine Learning Libraries	XGBoost, Scikit-learn, Pandas, NumPy
	Data Visualization	Matplotlib, Seaborn
	Development Environment	Jupyter Notebook, Visual Studio Code, PyCharm
	Version Control	Git, GitHub
Data	Dataset Source	Kaggle Flood Prediction Dataset, Government Weather & Rainfall Data
	Data Format	CSV
	Model	XGBoost Classifier
	Deployment	Flask Web Application

Expected Outcome:

The **Rising Waters** system will provide accurate flood predictions, real-time risk assessment, and early warning alerts by leveraging machine learning and weather data. The solution will assist disaster management authorities, emergency response teams, and citizens in making timely decisions, reducing flood-related damage, improving public safety, and enhancing overall disaster preparedness.